Newton–Cotes Integration Rules — Composite Formula Cheatsheet

Trapezoidal Rule (n=1)

$$I = \frac{h}{2}[y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

Explanation: Approximates f(x) with straight lines between each pair of consecutive points. Each interval forms a trapezoid. The area under each trapezoid is h/2 × (sum of two heights). Interior points are counted twice (coefficient 2) because they're shared between adjacent trapezoids.

When to use: When you need a quick, simple approximation. O(h^2) accuracy.

where: - h = (b - a)/n - n = number of intervals - y_i = f(x_i)

Simpson 1/3 Rule (n=2)

$$I = \frac{h}{3}[y_0 + 4(y_1 + y_3 + y_5 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2}) + y_n]$$

Explanation: Approximates f(x) with parabolas (quadratic polynomials) fitted through every 3 consecutive points. Every pair of intervals (h = (b-a)/2 for each pair) uses the single-panel formula h/3(y + 4y + y). The "1/3" name comes from the h/3 coefficient. Odd-indexed interior points get coefficient 4 (they're at the peaks of parabolas), even-indexed get coefficient 2 (they're shared between adjacent parabolas).

When to use: Best balance of accuracy and simplicity. O(h) accuracy. Very popular!

where: - h = (b - a)/n - **n must be EVEN** - Odd indices (1, 3, 5, …, n-1) have coefficient 4 - Even indices (2, 4, 6, …, n-2) have coefficient 2 - Endpoints (y , y) have coefficient 1

Simpson 3/8 Rule (n=3)

$$I = \frac{3h}{8}[y_0 + 3(y_1 + y_2 + y_4 + y_5 + y_7 + y_8 + \cdots) + 2(y_3 + y_6 + y_9 + \cdots) + y_n]$$

Or equivalently:

$$I = \frac{3h}{8}[y_0 + 3\sum_{i\equiv 1,2 \pmod 3} y_i + 2\sum_{i\equiv 0 \pmod 3, i>0, i<n} y_i + y_n]$$

Explanation: Approximates f(x) with cubic polynomials (degree 3) fitted through every 4 consecutive points. Every group of 3 intervals uses formula 3h/8(y + 3y + 3y + y). The "3/8" name comes from the coefficient. The weight pattern repeats every 3 intervals: points at positions (i mod 3) = 1,2 get coefficient 3; points at (i mod 3) = 0 (except endpoints) get coefficient 2.

When to use: When n must be divisible by 3. Also O(h) accurate like Simpson 1/3, but different weight distribution.

where: - h = (b - a)/n - **n must be MULTIPLE OF 3** - Indices where (i mod 3) = 1 or 2: coefficient 3 - Indices where (i mod 3) = 0 (except 0 and n): coefficient 2 - Endpoints (y , y): coefficient 1

Boole's Rule (n=4)

$$I = \frac{2h}{45}[7y_0+32(y_1+y_5+y_9+\cdots)+12(y_2+y_6+y_{10}+\cdots)+32(y_3+y_7+y_{11}+\cdots)+7(y_4+y_8+y_{12}+\cdots)+y_n]$$

Or more clearly:

$$I = \frac{2h}{45}[7y_0+32\sum_{i\equiv 1 \pmod 4} y_i+12\sum_{i\equiv 2 \pmod 4} y_i+32\sum_{i\equiv 3 \pmod 4} y_i+7\sum_{i\equiv 0 \pmod 4, i>0} y_i+y_n]$$

Explanation: Approximates f(x) with quartic polynomials (degree 4) through 5 consecutive points. Every 4 intervals uses single formula 2h/45(7y + 32y + 12y + 32y + 7y). The weights (7, 32, 12, 32) repeat, giving maximum weight 32 to the interior points. Much higher accuracy than Simpson's rules!

When to use: When you need better accuracy and n is divisible by 4. O(h) accuracy — excellent!

Weight pattern repeats every 4 intervals: 7, 32, 12, 32

where: - h = (b - a)/n - **n must be MULTIPLE OF 4** - Coefficient pattern: 7, 32, 12, 32, repeating - Last y_n gets coefficient 7

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n=5 Rule (6-point Closed Newton–Cotes)

$$I = \frac{5h}{288}[19y_0+75(y_1+y_6+y_{11}+y_{16}+\cdots)+50(y_2+y_3+y_7+y_8+y_{12}+y_{13}+\cdots)+75(y_4+y_9+y_{14}+\cdots)+y_n]$$

Weight pattern (repeats every 5 intervals): 19, 75, 50, 50, 75

Explanation: Uses degree-5 polynomials fitted through 6 consecutive points. The weight pattern repeats every 5 intervals: position 0 → 19, position 1 → 75, position 2 → 50, position 3 → 50, position 4 → 75, then repeats at position 5 (becomes position 0 of next group). So y , y , y ... all get coefficient 75, and y , y , y , y ... all get coefficient 50.

When to use: When n is divisible by 5 and you need very high accuracy. O(h) convergence.

where: - h = (b - a)/n - **n must be MULTIPLE OF 5** - Coefficient pattern repeats: 19, 75, 50, 50, 75, repeating every 5 points - y , y , y ... get 75 - y , y , y , y ... get 50 - y , y , y ... get 75

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n=6 Rule (7-point Closed Newton–Cotes)

$$I = \frac{h}{140}[41y_0+216(y_1+y_7+y_{13}+\cdots)+27(y_2+y_6+y_8+y_{12}+\cdots)+272(y_3+y_9+y_{15}+\cdots)+27(y_4+y_{10}+\cdots)+216(y_5+y_{11}$$

Weight pattern (repeats every 6 intervals): 41, 216, 27, 272, 27, 216

Explanation: The most accurate closed Newton–Cotes rule using degree-6 polynomials through 7 consecutive points. The weight pattern repeats: position 0 → 41, position 1 → 216, position 2 → 27, position 3 → 272 (maximum!), position 4 → 27, position 5 → 216, then repeats. Notice the **symmetric pattern** (41...216...27...272...27...216) that emphasizes the center position 3 with weight 272. This gives exceptional accuracy.

When to use: When absolute maximum accuracy is required and n is divisible by 6. O(h) convergence — highest accuracy among closed Newton–Cotes!

where: - h = (b - a)/n - **n must be MULTIPLE OF 6** - Coefficient pattern repeats: 41, 216, 27, 272, 27, 216, then repeats - Center position always gets 272 (maximum weight)

Quick Reference Table

Rule	Coefficient	Divisor	Pattern (repeating)	n Multiple
Trapezoidal	h/2	1	1, 2, 2, ..., 2, 1	Any
Simpson 1/3	h/3	1	1, 4, 2, 4, 2, ..., 4, 1	Even
Simpson 3/8	3h/8	1	1, 3, 3, 2, 3, 3, 2, ..., 1	÷3
Boole	2h/45	1	7, 32, 12, 32, ..., 7	÷4
n=5	5h/288	1	19, 75, 50, 50, 75, ...	÷5
n=6	h/140	1	41, 216, 27, 272, 27, 216, ...	÷6

Worked Solutions with Examples

Example n=5: ^{1} $1/(x^2+1)$ dx using n=5 Rule with n=10 intervals

Step 1: h = (1 - 0)/10 = 0.1

Step 2: Calculate y values at x = 0, 0.1, 0.2, ..., 1.0: - y = 1.0000 - y = 0.9901 - y = 0.9615 - y = 0.9174 - y = 0.8621 - y = 0.8000 - y = 0.7353 - y = 0.6711 - y = 0.6098 - y = 0.5525 - y = 0.5000

Step 3: Group by weight pattern (19, 75, 50, 50, 75, repeating): - Position 0 (y): coefficient 19 → 19(1.0000) = 19.0000 - Position 1 (y , y): coefficient 75 → 75(0.9901 + 0.7353) = 75(1.7254) = 129.405 - Position 2 (y , y): coefficient 50 → 50(0.9615 + 0.6711) = 50(1.6326) = 81.630 - Position 3 (y , y): coefficient 50 → 50(0.9174 + 0.6098) = 50(1.5272) = 76.360 - Position 4 (y , y): coefficient 75 → 75(0.8621 + 0.5525) = 75(1.4146) = 106.095 - Position 5 (y , y): coefficient 75 → 75(0.8000 + 0.5000) = 75(1.3000) = 97.500

Step 4: Apply formula:

$$I \approx \frac{5 \times 0.1}{288}[19.0 + 129.405 + 81.630 + 76.360 + 106.095 + 97.500]$$

$$= \frac{0.5}{288}[509.990]$$

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$$= 1.7708...$$

Wait, this needs recalculation. Let me recalculate correctly:

The pattern repeats every 5 intervals. With n=10 intervals, we have positions 0-10 (11 points): - Position 0 mod 5 = 0 (y): 19 - Position 1 mod 5 = 1 (y , y): 75 - Position 2 mod 5 = 2 (y , y): 50 - Position 3 mod 5 = 3 (y , y): 50 - Position 4 mod 5 = 4 (y , y): 75 - Position 5 mod 5 = 0 (y): 19 - Position 6 mod 5 = 1 (y): 75... but y is endpoint!

For composite rules, endpoints typically use their pattern value. Let me use the correct approach:

$$I \approx \frac{5h}{288}[19y_0 + 75(y_1 + y_6) + 50(y_2 + y_3 + y_7 + y_8) + 75(y_4 + y_9) + 19y_5 + \text{next cycle}]$$

Actually, with 11 points and repeating pattern, we apply it carefully:

$$I \approx \frac{0.5}{288}[19(1.0) + 75(0.9901 + 0.7353) + 50(0.9615 + 0.9174 + 0.6711 + 0.6098) + 75(0.8621 + 0.5525) + 19(0.8)]$$

Wait, let me reconsider. With n=10, the last point y should get treated. Let me use n=5 with 6 points instead for clarity:

Step 1: h = (1 - 0)/5 = 0.2

Step 2: Calculate y values at x = 0, 0.2, 0.4, 0.6, 0.8, 1.0: - y = 1.0000 - y = 0.9615 - y = 0.8621 - y = 0.7353 - y = 0.6098 - y = 0.5000

Step 3: Apply n=5 formula with 6 points:

$$I \approx \frac{5(0.2)}{288}[19y_0 + 75y_1 + 50y_2 + 50y_3 + 75y_4 + 19y_5]$$

$$= \frac{1}{288}[19(1.0) + 75(0.9615) + 50(0.8621) + 50(0.7353) + 75(0.6098) + 19(0.5)]$$

$$= \frac{1}{288}[19 + 72.1125 + 43.105 + 36.765 + 45.735 + 9.5]$$

$$= \frac{226.2175}{288}$$

$$= 0.78543$$

Error: | /4 - 0.78543| = |0.785398 - 0.78543| 0.000032

Example n=6: ¹ 1/(x²+1) dx using n=6 Rule with n=6 intervals

Step 1: h = (1 - 0)/6 = 1/6 0.16667

Step 2: Calculate y values at x = 0, 1/6, 2/6, 3/6, 4/6, 5/6, 1: - y = 1.0000 - y = 1/(1/36 + 1) = 36/37 0.97297 - y = 1/(4/36 + 1) = 1/(1/9 + 1) = 9/10 = 0.90000 - y = 1/(9/36 + 1) = 1/(1/4 + 1) = 4/5 = 0.80000 - y = 1/(16/36 + 1) = 1/(4/9 + 1) = 9/13 0.69231 - y = 1/(25/36 + 1) = 36/61 0.59016 - y = 0.5000

Step 3: Apply n=6 formula (weight pattern: 41, 216, 27, 272, 27, 216):

$$I \approx \frac{h}{140}[41y_0 + 216y_1 + 27y_2 + 272y_3 + 27y_4 + 216y_5 + 41y_6]$$

$$= \frac{1/6}{140}[41(1.0)+216(0.97297)+27(0.9)+272(0.8)+27(0.69231)+216(0.59016)+41(0.5)]$$

$$= \frac{1}{840}[41 + 210.1752 + 24.3 + 217.6 + 18.6923 + 127.4346 + 20.5]$$

$$= \frac{1}{840}[659.7921]$$

$$= 0.78546$$

Error: | /4 - 0.78546| = |0.785398 - 0.78546| 0.000062

How to Use (Step-by-Step)

Example: ^{1} $1/(x^2+1)$ dx using Simpson 1/3 with n=4 intervals

Step 1: h = (1 - 0)/4 = 0.25

Step 2: Calculate y values at x = 0, 0.25, 0.5, 0.75, 1: - y = 1 - y = $1/(0.25^2 + 1)$ = 0.9412 - y = $1/(0.5^2 + 1)$ = 0.8 - y = $1/(0.75^2 + 1)$ = 0.64 - y = $1/(1^2 + 1)$ = 0.5

Step 3: Identify odd and even indices: - Odd (1, 3): y = 0.9412, y = 0.64 - Even (2): y = 0.8

Step 4: Apply Simpson 1/3:

$$I = \frac{0.25}{3}[1 + 4(0.9412 + 0.64) + 2(0.8) + 0.5]$$

$$= \frac{0.25}{3}[1 + 4(1.5812) + 1.6 + 0.5]$$

$$= \frac{0.25}{3}[1 + 6.3248 + 1.6 + 0.5]$$

$$= \frac{0.25}{3}[9.4248]$$

$$= \frac{2.3562}{3} = 0.7854$$

Summary: What to Remember for Exam

1. **Trapezoidal:** All interior points get coefficient 2
2. **Simpson 1/3:** Odd points get 4, even points get 2
3. **Simpson 3/8:** Repeating pattern 1, 3, 3, 2
4. **Boole:** Repeating pattern 7, 32, 12, 32
5. **Higher n:** Use repeating weight patterns

Key: h = (b - a)/n, multiply all values by h/(divisor)